Short Communication

Dietary Pattern and Diet Diversity of Undergraduate Students of Eastern India: An Online Survey Based Cross-Sectional Study

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Abstract

Dietary diversity is a key component of a high-quality diet and is associated with reduced risks of chronic diseases and undernutrition. Recognizing its importance, this study aimed to identify the dietary patterns and diet diversity scores among undergraduate students in Eastern India. A cross-sectional, online questionnaire-based study was conducted over six months. Data were collected through a Google Form shared via social media and personal contacts, with informed consent obtained from all participants. The questionnaire included sections on general information, dietary variety and patterns, and anthropometric details. Ethical approval was obtained from the Institutional Ethics Committee of MTMC, Jamshedpur, and data analysis was performed using Jamovi software (version 2.2.5). A total of 354 students participated, most of whom were MBBS students from private colleges in Jharkhand and residing in hostels. Although most students reported consuming regular meals and snacks, breakfast skipping (74.8%) and irregular mealtimes (75.2%) were highly prevalent. Male students and those enrolled in MBBS programs demonstrated higher Diet Diversity Scores, which showed a significant association with Body Mass Index (BMI), suggesting a link between dietary diversity and nutritional status. In conclusion, the study highlights the need for improving diet diversity scores among undergraduate students in Eastern India. Furthermore, the observed irregular eating patterns and frequent breakfast skipping call for increased awareness and interventions to promote healthier dietary habits among this population.

Keywords: Dietary pattern, diet diversity, Eastern India, undergraduate students

Introduction

Dietary diversity (DD) has consistently been recognized as an important component of high-quality diets. Local and international dietary standards advise increasing food diversity since dietary factors are linked to an increased risk of chronic diseases and undernutrition.[1] The phrase "number of foods or food types ingested over a certain reference period" has been used to define dietary diversity.[2] Dietary Diversity Score (DDS) is a measure of diet quality that has been shown to be negatively correlated with the likelihood of developing chronic diseases.[3] Dietary patterns explain the entire diet, the foods, dietary groups, and nutrients included, their combinations and variety, as well as the regularity and amount of consumption.[4]

The National Institute of Nutrition (NIN) and the Indian Council of Medical Research (ICMR) have recommended a daily plate that ensures diet diversity, with only 8% of the plate being allocated to fruits and vegetables and only 45% to cereals, 17% to pulses, eggs, and meat, 12% to fats and oils, 10% to milk and curd, and 8% to nuts and seeds. The dietary diversity questionnaire serves as the quick, simple, and affordable diagnostic tool. It is simple to score and analyses the data gathered through the questionnaire. The food groups that a family or an individual ingested over the previous 24 hours are simply counted to determine the dietary variety scores outlined in these recommendations. The recommendations cover using the dietary diversity questionnaire both at the household and individual levels, where the calculation of the score varies slightly. The gathered information can also be examined to provide details on food groups of interest.[5] The ten food groups used in the study were selected from the booklet of Food and Agriculture Organization of the United Nations

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Rome, 2018 titled Dietary Assessment: A resource guide to method selection and application in low resource settings. The selected 10 food groups were all starchy staples, beans and peas, nuts and seeds, all dairy, flesh foods, eggs, vitamin A rich dark green leafy vegetables, other vitamin A rich vegetables and fruits, other vegetables, and other fruits.[6]

Calculating the diet score and obtaining information about the dietary pattern of undergraduate students across Eastern India will provide an overview of their nutritional status and can help them to promote healthy lifestyle. The study aims to identify the dietary pattern of undergraduate students of Eastern India and to know the DD and DDS of undergraduate college students using 24 hour recall method.

Methodology

a. **Study design:** Descriptive cross-sectional study.
b. **Study area:** To participate in this study, colleges in Eastern India that provide undergraduate programmed (MBBS, BDS, B.E./B. Tech, B.Sc., B.A., LLB, B. Commerce, and others) were contacted.
c. **Study setting:** Department of Community Medicine, Manipal Tata Medical College, Jamshedpur
d. **Study Duration:** 6 months
e. **Study Population:** Undergraduate students studying in various colleges in Eastern India as shown in Figure 1
   - **Exclusion criteria:** Students who did not give consent for the study.
f. **Sampling Technique:** Convenience snowball sampling method was used. The link of the online questionnaire was shared to the known contacts of a college and they were requested to spread the link to other students, and so the questionnaire was spread for data collection.
g. **Sample size:** The total number of respondents, who gave consent to participate in the study in the prescribed duration of data collection was 354. The participants filled the self-reported online questionnaire that was circulated to take part in the research as study participants.
h. **Data collection method:** Undergraduate students from different educational institutions located in Eastern India (Bihar, Jharkhand, West Bengal, Odisha, and Chhattisgarh) completed the questionnaires online by sharing the link with known personal contacts and contacts shared by the faculty in charge through social media platforms like Facebook, WhatsApp, and Instagram. The number of food categories consumed in the preceding 24 hours from the total food groups as stated by the Food and Agriculture Organization (FAO) was used to measure dietary diversity.[8] Dietary habits are tracked by asking a few key questions.
i. **Study Tool:** A pre-tested semi-structured survey form was created with a consent form added to it and was shared over the Google Forms platform to collect data. The questionnaire used in the research was validated by experts and the ten food groups consumption and their score were assigned using the reference from Minimum Dietary Diversity for women - A Guide to Measurement" (FAO, 2016). The questionnaire asked for general information, details about the variety and pattern of diets, and details about anthropometric measures. **i) General Information:** Age, Gender, Educational Stream, Location of College (state or union territory), Type of Institution and Place of Current Residence (Hostel, Home etc.) **ii) Questions on diet diversity and pattern:** Type of Diet, Number of major meals in a day, usual source of meals, timely consumption of meals, Skipping breakfast, snacking habits, consumption of the different food groups in last 24 hour. **iii) Anthropometric measurements as given by the participants:** Weight (in kg) and height (in cm) and Body Mass Index (BMI). **iv) Food groups used in dietary diversity assessment:** For calculation of DDS, one point is awarded, making it equally part of the final score, even if each food category has one or more highlighted foods. If none of the foods in the food group were eaten, the value is zero.[6] The total DDS is the sum of the dietary groups consumed**.** Ten food groups were used to assess the diet diversity score (DDS). The food groups were as. DDS Score was grouped into categories as Good (DDS ≥5), Medium (DDS = 3–4) and Low (DDS ≤2).
j. **Data cleaning and Statistical Analysis:** After collection of data the templates were generated in MS excel Sheets and jamovi (solid version 2.2.5) software (to determine chi square) was used for analysis of data. The threshold for statistical significance was $P < 0.05$.
k. **Ethical approval:** The study was presented before the Institutional Ethics Committee (IEC) of the Institution



Figure 1: Map showing selected state of eastern parts of India[7]

Table 1: General information of the study participants (*n*=354)

Variable	Category	Frequency	Percentage (%)	95% CI
Gender	Female	175	49.4%	0.49 (0.44 – 0.54)
	Male	179	50.6%	0.50 (0.45 – 0.55)
Type of Institution	Government	155	43.7%	0.43 (0.38 – 0.48)
	Private	199	56.2%	0.56 (0.51 – 0.61)
Stream	MBBS	271	76.3%	0.76 (0.71 – 0.80)
	BDS	15	4.2%	0.04 (0.02 – 0.06)
	B.A	5	1.4%	0.01 (0.006 – 0.03)
	B. Commerce	7	2.1%	0.19 (0.009 – 0.04)
	B.E./ B. Tech.	24	7%	0.06 (0.04 – 0.09)
	Others*	32	9%	0.09 (0.06 – 0.14)
Currently Residing In	Hostel	314	88.7%	0.88 (0.84 – 0.91)
	Paying guest	5	1.4%	0.014 (0.006 – 0.03)
	Home	25	7.06%	0.07 (0.04 – 0.10)
	Rented Flats	10	2.82%	0.02 (0.01 – 0.05)
Location of the institute	Bihar	82	23.16%	0.23 (0.19 – 0.27)
	Jharkhand	221	62.4%	0.62 (0.57 – 0.67)
	Odisha	38	10.7%	0.10 (0.07 – 0.14)
	West Bengal	11	3.1%	0.03 (0.01 – 0.05)
	Others	2	0.56%	0.005 (0.0016 – 0.020)

*Others include the streams of BCA, BBA and those not specified by the respondent

Table 2: Distribution of Mean DDS and Standard Deviation across various undergraduate streams using Analysis of Variance (ANOVA)

Stream	Frequency	Mean	SD	*P*
MBBS	271	3.52	1.95	0.0017
BDS	15	3.01	1.78	
B.E./ B. Tech.	24	3.25	1.91	
B. A.	5	3.24	1.45	
B. Commerce	7	2.51	1.31	
Others	32	3.10	2.09	

and approval was provided (IEC Ref. No: MTMC/IEC/2023/29)

Results

Table 1: Among the 354 participants, gender distribution was nearly equal (Male: 50.6%, Female: 49.4%). The majority were from private institutions (56.2%) and enrolled in MBBS courses (76.3%). Most students resided in hostels (88.7%), with the highest concentration of institutes located in Jharkhand (62.4%).

Table 2: DDS varied significantly across undergraduate streams (p = 0.0017), with MBBS students having the highest mean DDS (3.52 ± 1.95) and B. Commerce students the lowest (2.51 ± 1.31).

Table 3: A statistically significant association was found between DDS category and BMI class ($P = 0.002$). Participants with a good DDS were more likely to have normal BMI, while low DDS was associated with higher prevalence of underweight and obesity.

Table 4: The mean Diet Diversity Score (DDS) among all participants was 3.5 (SD = 1.7), indicating moderate dietary diversity.

Discussions

The study involved 354 participants with nearly equal gender representation. MBBS stream students constituted the majority, likely due to the study's initiation within a medical college. Most participants resided in hostels, with a small number living in paying guest accommodations or at home. A study by Paraparaumu indicates that household diet diversity serves as a proxy for diet quality and nutritional satisfaction. DD is positively linked to food security's pillars: Availability, access, and utilization. A multiple regression analysis showed that higher purchasing power, larger families, and more educated male heads of households correlated with higher dietary diversity scores. Access to the Public Distribution System (PDS) indirectly boosted dietary diversity by making a wider range of foods more affordable. Conversely, lower social standing, indicated by Scheduled Caste/Tribe (SC/ST) membership, was associated with reduced diversity scores. Policy interventions aimed at improving diet and nutrition security should therefore consider the socioeconomic context of the target population.[9]

A study by Kingsley Omage *et al.*[10] found that the mean age of respondents was 23.5 ± 2.4 years, with a higher proportion of women participating. Over half of the respondents skipped breakfast and ate between meals. Breakfast skipping was more prevalent among females, with a significant correlation ($P < 0.010$) between gender and skipping breakfast. Snacking was common, and there was a significant correlation ($P < 0.034$) between age group and

Table 3: Association of DDS Category with different BMI classes among the study participants

Category	Category of BMI									
	Underweight (%)		Normal (%)		Overweight (%)		Obese (%)		Total	*P*
Good	8	(24.6.%)	84	(39.8%)	30	(39%)	9	(26.5%)	131	0.002
Medium	14	(43.2%)	71	(33.7%)	25	(32.1%)	14	(40.6%)	124	
Low	10	(32.2%)	55	(26.5%)	23	(28.9%)	11	(32.9%)	99	
Grand Total	32	(100%)	210	(100%)	78	(100%)	34	(100%)	354	

Table 4: Distribution of Mean DDS and Standard Deviation across gender using unpaired *t*-test

Gender	Frequency	Mean	SD	*P*
Male	179	3.46	2.02	0.0001
Female	175	3.12	1.83	

snacking status. Males exhibited a significantly higher dietary diversity score ($P < 0.004$) than females. The study concluded that skipping breakfast and eating between meals are common and recommended the implementation of a regular nutrition education program focusing on appropriate eating behaviors.

A study by Kumar *et al.*[11] found participants had a mean age of 20.7 ± 1.9 years, with a majority being female. DDS indicated 34.7% were low and 28.5% were high. Males exhibited the higher mean DDS than females, with an overall mean DDS of 3.4. Among individuals with good DDS, 57.1% ate from a mess, and 63.1% had a normal BMI. DDS showed significant variation across geographies ($P = 0.0007$), graduating streams ($P = 0.0014$), and meal sources ($P < 0.001$). Approximately 75% of participants demonstrated insufficient diet diversification, highlighting poor eating patterns and unsuitable food consumption behaviors. According to a study by Vibhute NA *et al.*,[12] there needs to be more research conducted on the nutritional aspects of our future medical professionals' diets, and prompt interventions should be launched to ensure that the WHO's guidelines are understood and applied.

Strength and limitations

The study highlights snowball sampling's advantages for data collection across broad geographic areas, especially when formal sample frames are absent, facilitating access to diverse participants and potentially capturing regional dietary variations. However, this non-probability method introduces selection bias and limits generalizability to the wider student population. Potential uniformity within participant networks may also affect the variety of food patterns recorded. Despite these limitations, the research offers valuable preliminary insights into dietary variations among Eastern Indian undergraduates, informing future, more comprehensive studies.

Conclusions

The study concluded that participants had a mean age of 20.7 ± 1.9 years. Their mean DDS was 3.5, indicating less than good dietary diversity and improper food consumption habits, suggesting poor eating patterns among undergraduate students. The study offered an overview of these dietary patterns and diversity and recommended further research on the larger population for more generalizable results.

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Nil.

Conflicts of interest

There are no conflicts of interest.

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